

Brenier's Theorem

Assumptions:

1. $c(x, y) = \|x - y\|^2$
2. α has a density $\rho_\alpha = \frac{d\alpha}{dx}$



Monge's Problem:

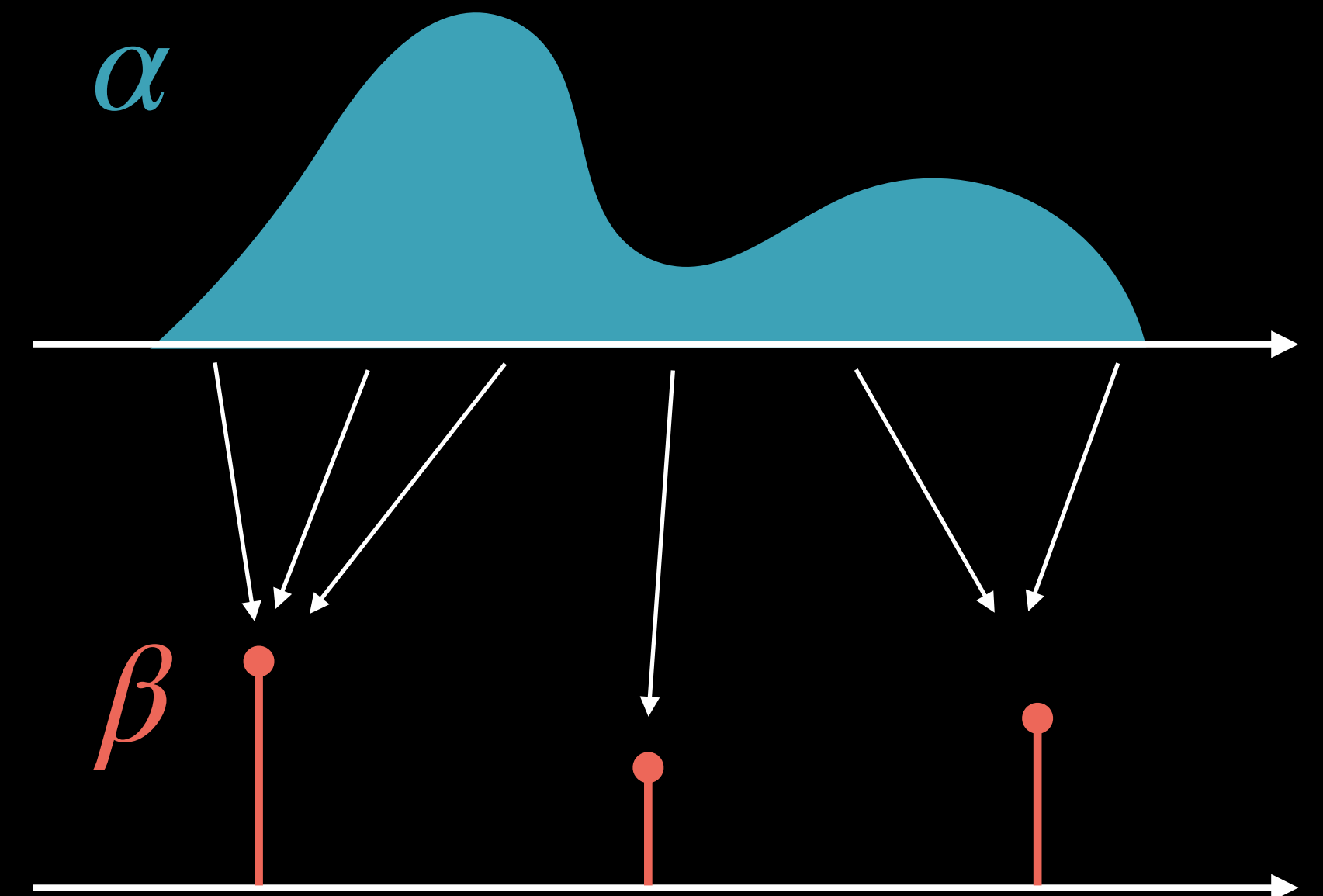
(Continuous / general version)

$$\min_{\beta=T_{\#}\alpha} \int_X c(x, T(x)) d\alpha(x)$$

Theorem [Brenier, 1991]:

1. There exists a **unique** map T that solves the Monge problem
2. T is the gradient of a convex function

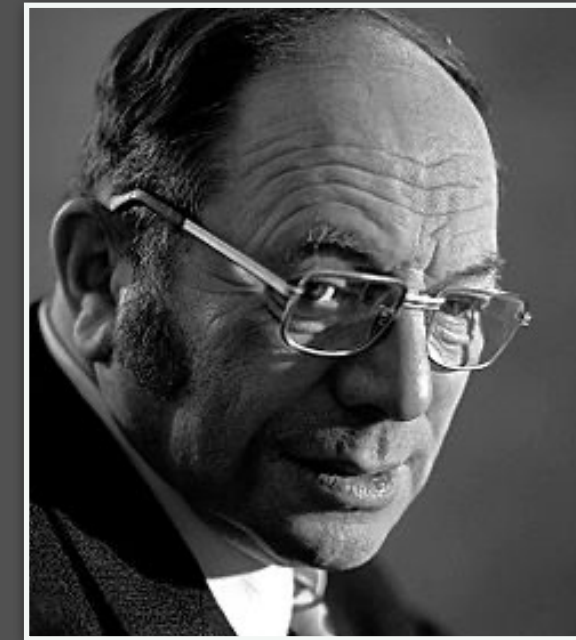
Formally: the unique $T = \nabla \varphi$ such that φ is **convex** and $(\nabla \varphi)_{\#}\alpha = \beta$



Brenier's Theorem

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Kantorovich's Problem:

(Continuous version)

$$\min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) d\pi(x, y)$$

